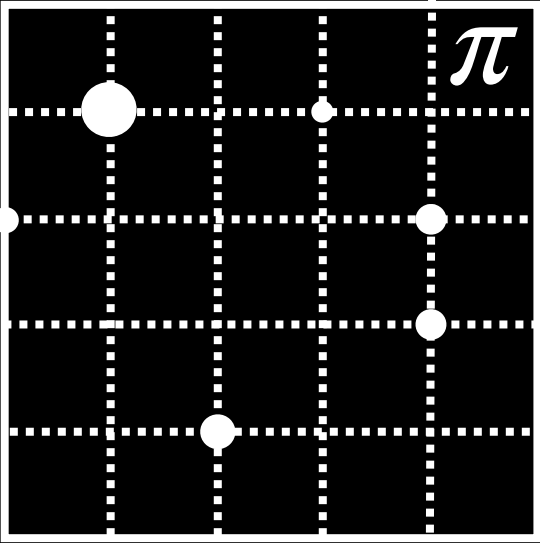




b_1 b_2 $\dots$ b_m β



A diagram showing a grid of points. The grid is defined by dashed lines. There are four columns and four rows. The columns are labeled at the top with b_1 , b_2 , $\dots$, b_m , and β . The rows are labeled on the left with a_1 , a_2 , $\vdots$, and a_n . The label α is placed below the row labels. The label π is placed in the top right corner of the grid. Points are marked as follows: a large white point at the intersection of the first column and the first row; a small white point at the intersection of the third column and the first row; a small white point at the intersection of the first column and the second row; a small white point at the intersection of the fourth column and the second row; a small white point at the intersection of the fourth column and the third row; and a large white point at the intersection of the second column and the fourth row. The points b_1 , b_2 , and b_m are represented by red circles of varying sizes, with b_1 being the largest and b_2 and b_m being smaller. The point β is represented by a red circle of the same size as b_2 and b_m . The point α is represented by a teal circle of the same size as a_1 .

a_1

a_2

$\vdots$

a_n

α

π



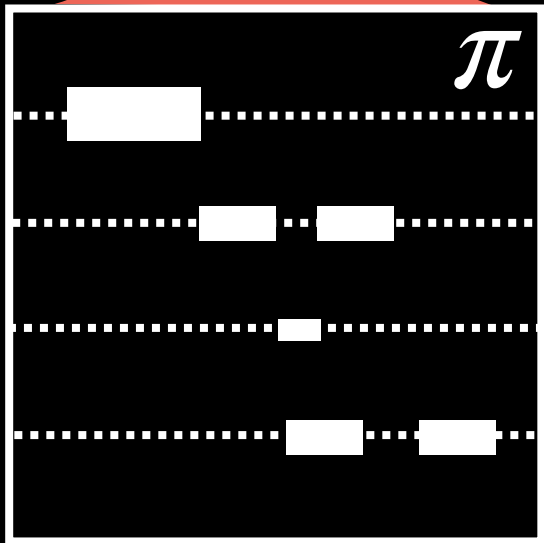
a_1 ●

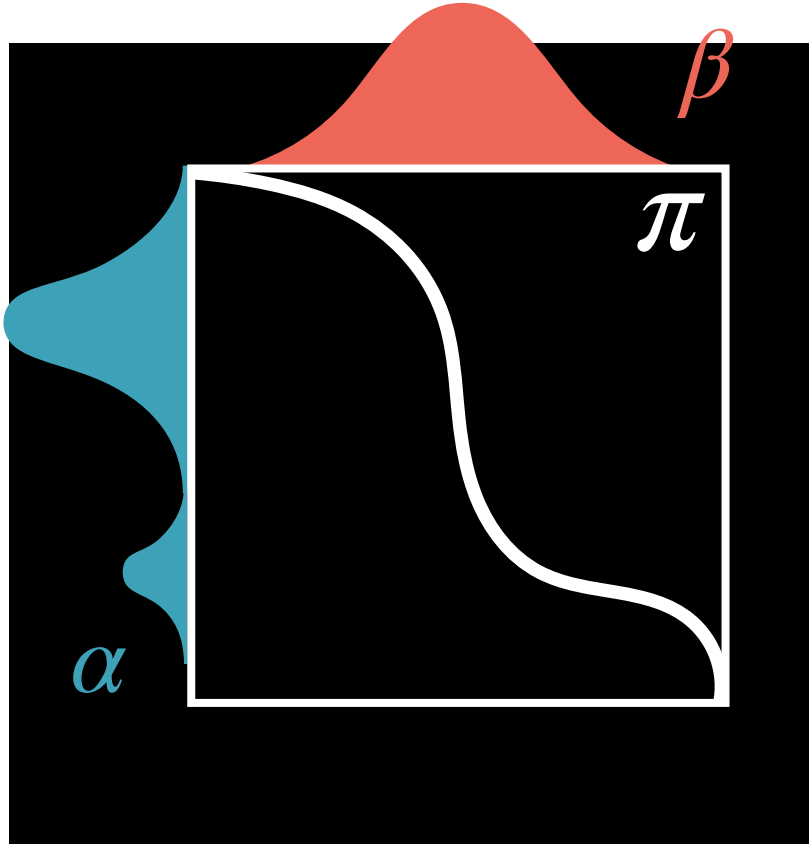
a_2 ●

$\vdots$ ●

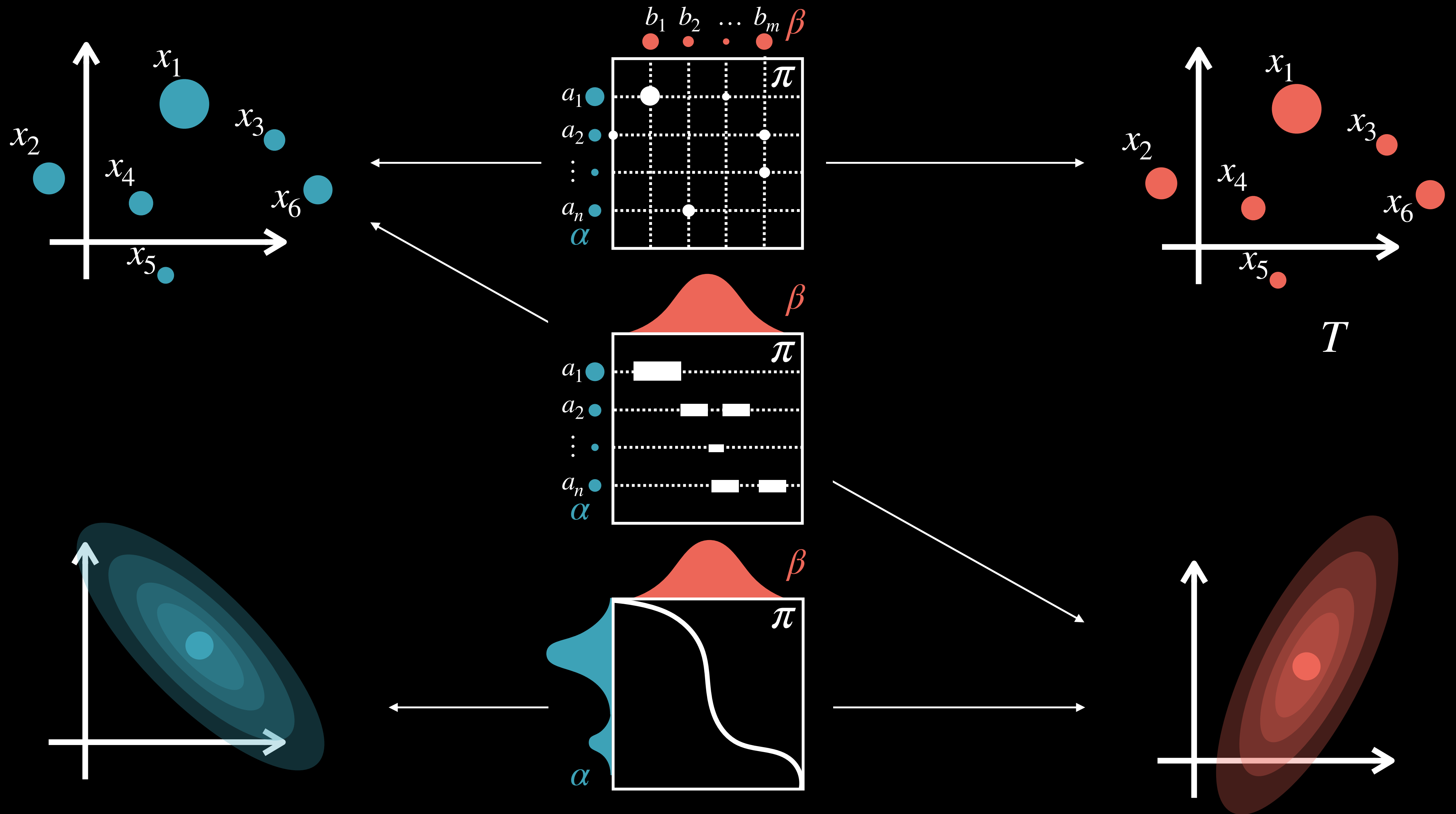
a_n ●

α





Three types of couplings



Re-visiting Brenier's Theorem